

Core Builds

Stream Expression Reference

ESEs · PSEs · ISEs · LABS — what they do and why

v2.9.x / LABS v0.9.0 · June 2026

About this document

AIOStreams evaluates three types of stream expressions against every result before presenting streams to the user. This reference documents every expression in the Core Builds template suite — what problem it solves, the logic behind the implementation, and which templates it appears in. LABS expressions are additional features in testing in the Nightly template tier.

Type	Tag	Effect & order
Exclusion Stream Expression	ESE	Removes matching streams from the pool. Runs first.
Priority Stream Expression	PSE	Reorders remaining streams by quality tier, bitrate, codec, audio. Runs second.
Inclusion Stream Expression	ISE	Forces specific streams back in regardless of ESE exclusions. Runs last.
LABS expression	LABS	Experimental expression in Nightly templates — not yet in stable release.

Exclusion Stream Expressions (ESE)

ESEs run first and permanently remove streams from the pool. An ESE that returns streams causes those streams to be excluded; an ESE returning `[]` excludes nothing. Evaluation order: flood control → hard quality/type kills → Tamtaro synced set → content-aware guards (seeders, scores, anime) → deduplication → final limit → CB kills.

ESE Per-Addon Flood Guard

Used in: All active templates with multiple scrapers

Hard caps on how many non-library, non-SeaDex cached results each scraper can contribute. Prevents any single indexer dominating on popular titles. Caps (v2.9.x): Meteor ≤5, Comet ≤5, MediaFusion ≤4, EZTV ≤3, HdHub ≤3, Torrent Galaxy ≤1, Knaben ≤1, TorrentsDB ≤1. Library and SeaDex streams are exempt and always pass through.

```
merge(slice(addon(negate(merge(library(streams),seadex(streams)),cached(streams)), 'Meteor'),5),slice(addon(negate(merge(library(streams),seadex(streams)),cached(streams)), 'Comet RD'),5),slice(addon(negate(merge(library(streams),seadex(streams)),cached(streams)), 'MediaFusion'),4),slice(addon(negate(merge(library(streams),seadex(streams)),cached(streams)), 'Torrent Galaxy'),1),slice(addon(negate(merge(library(streams),seadex(streams)),cached(streams)), 'EZTV'),3),slice(addon(negate(merge(library(streams),seadex(streams)),cached(streams)), 'Knaben'),1),slice(addon(negate(merge(library(streams),seadex(streams)),cached(streams)), 'HdHub'),3),slice(addon(negate(merge(library(streams),seadex(streams)),cached(streams)), 'TorrentsDB'),1))
```

ESE Usenet Propagation Guard

Used in: All templates with Usenet presets

Drops Usenet NZBs posted less than 2 hours ago (0.083 days). Usenet articles take time to propagate across all backbone servers — streaming a brand-new NZB often hits incomplete articles mid-download. Only fires when older Usenet results exist alongside the new ones.

```
count(negate(age(type(streams,'usenet','stremio-usenet'),0,'0.083'),type(streams,'usenet','stremio-usenet'))>0?age(type(streams,'usenet','stremio-usenet'),0,'0.083'):[[]])
```

ESE AI Upscale Exclusion

Used in: All active templates

Keyword-kills streams where the filename contains AI upscaling markers: topaz, ai-upscale, aiupscale, upscaled, neural, enhancedai. AI upscales inflate file sizes while delivering worse quality than native encodes at the same bitrate. Library and SeaDex are exempt.

```
keyword(negate(merge(library(streams),seadex(streams)),streams),'all','topaz','ai-upscale','aiupscale','upscaled','neural','enhancedai')
```

ESE Hard Resolution Kill — Stream is 1080p-only

Used in: 1080p templates (Stream, Essential, Hybrid, EasyNews 1080p)

Hard-excludes 4K (2160p, 1440p) and sub-1080p streams (720p, 576p, 480p). 1080p templates have no 4K PSE tiers so 4K results would sort incorrectly. Keeps the result set clean and sort order meaningful.

```
resolution(streams, '2160p', '1440p', '720p', '576p', '480p')
```

ESE Part of Tamtaro SEL Setup

Used in: All active templates

Placeholder ESE — grouping marker for the Tamtaro synced ESE block. Returns `[]` always. The actual logic loads via syncedExcludedStreamExpressionUrls.

```
[[]]
```

ESE Standard ESE v1.2.8 [git.tamtaro.de]

Used in: All active templates

Version label for Tamtaro's synced exclusion set. Returns `[]` inline. Kept for display purposes in the AIOStreams UI.

```
[[]]
```

ESE G's Low Bitrate

Used in: All active templates

Excludes streams with abnormally low bitrate relative to their resolution. Catches heavily compressed re-encodes that pass quality/resolution checks but deliver visibly degraded playback.

```
negate((isAnime or 'Animation' in
genres?bitrate(streams,1,'0.67Mbps'):merge(bitrate(quality(resolution(streams,'2160p'),'Bluray REMUX'),1,'25Mbps'),bitrate(quality(resolution(streams,'2160p'),'Bluray'),1,'13.6Mbps'),bitrate(quality(resolution(streams,'2160p'),'WEB-DL','WEBRip'),1,'4.6Mbps'),bitrate(quality(resolution(streams,'2160p'),'HDTV'),1,'11.33Mbps'),bitrate(quality(resolution(streams,'1080p'),'Bluray REMUX'),1,'13.6Mbps'),bitrate(quality(resolution(streams,'1080p'),'Bluray'),1,'6.77Mbps'),bitrate(quality(resolution(streams,'1080p'),'WEB-DL','WEBRip'),1,'1.67Mbps'),bitrate(quality(resolution(streams,'1080p'),'HDTV'),1,'4.51Mbps'),bitrate(quality(resolution(streams,'720p'),'Bluray'),1,'3.43Mbps'),bitrate(quality(resolution(streams,'720p'),'WEB-DL','WEBRip'),1,'1.67Mbps'),bitrate(quality(resolution(streams,'720p'),'HDTV'),1,'2.28Mbps'),bitrate(streams,1,'0.67Mbps'))),cached(streams))>10?(isAnime or 'Animation' in genres?bitrate(streams,1,'0.67Mbps'):merge(bitrate(quality(resolution(streams,'2160p'),'Bluray REMUX'),1,'25Mbps'),bitrate(quality(resolution(streams,'2160p'),'Bluray'),1,'13.6Mbps'),bitrate(quality(resolution(streams,'2160p'),'WEB-DL','WEBRip'),1,'4.6Mbps'),bitrate(quality(resolution(streams,'2160p'),'HDTV'),1,'11.33Mbps'),bitrate(quality(resolution(streams,'1080p'),'Bluray REMUX'),1,'13.6Mbps'),bitrate(quality(resolution(streams,'1080p'),'Bluray'),1,'6.77Mbps'),bitrate(quality(resolution(streams,'1080p'),'WEB-DL','WEBRip'),1,'1.67Mbps'),bitrate(quality(resolution(streams,'1080p'),'HDTV'),1,'4.51Mbps'),bitrate(quality(resolution(streams,'720p'),'Bluray'),1,'3.43Mbps'),bitrate(quality(resolution(streams,'720p'),'WEB-DL','WEBRip'),1,'1.67Mbps'),bitrate(quality(resolution(streams,'720p'),'HDTV'),1,'2.28Mbps'),bitrate(streams,1,'0.67Mbps'))):[]
```

ESE ongoingSeasonPack

Used in: All active templates

For currently-airing series: excludes season packs when the season is not yet complete. A season pack grabbed mid-season would be missing episodes. Checks ongoingSeason, daysSinceLastAired, and queryType before firing.

```
((queryType=='series' or queryType=='anime.series') and ongoingSeason and (daysSinceLastAired <-1 or daysUntilNextEpisode>=0))?seasonPack(streams,'onlySeasons'):[]
```

ESE Low Seeders

Used in: All active templates

Excludes torrent streams below a minimum seeder threshold. Does not apply to cached streams (cached torrents need no active seeders) or Usenet streams.

```
count(seeders(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),2))<=5 ?[]:count(seeders(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),q2(values(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),seeders))))>20?seeders(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),0,max(1,q2(values(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),seeders)))):count(seeders(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),q1(values(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),seeders))))>20?seeders(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),0,max(1,q1(values(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),seeders)))):count(seeders(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),percentile(values(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),seeders),10)))>20?seeders(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),0,max(1,percentile(values(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),seeders),10)))>20?seeders(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),2))>5?negate(seeders(merge(type(streams,'p2p'),type(uncached(streams),'debrid')),1),merge(type(streams,'p2p'),type(uncached(streams),'debrid'))):[]
```

ESE Extra SeaDex

Used in: All active templates

Deduplication: removes redundant SeaDex results so each release appears at most once. SeaDex streams are already boosted by PSEs/ISEs — duplicates from multiple scrapers would otherwise clutter the results.

```
count(seadex(streams,'best'))>1 or count(negate(seadex(streams,'best'),seadex(streams)))>1?merge(slice(negate(seadex(streams,'best'),seadex(streams)),1),slice(seadex(streams,'best'),1)):[]
```

ESE Bad 4k Anime

Used in: All active templates

Excludes 4K anime streams known to be problematic — typically upscales or re-encodes with inflated resolution. Most anime is mastered at 1080p or lower.

```
(isAnime and originalLanguage == 'Japanese' and count(quality(resolution(cached(streams), '2160p'), 'Bluray REMUX')) == 0 and count(seadex(resolution(streams, '2160p'))) == 0)? negate(merge(library(streams), seadex(streams)), resolution(streams, '2160p')) : []
```

ESE Upscaled 4k

Used in: All active templates

Excludes 4K streams with a visual upscale tag — where the source master is 1080p or lower, algorithmically upscaled to 2160p. Distinct from the keyword-based AI Upscale Exclusion; this uses structured visualTag() metadata.

```
(queryType=='movie' or queryType=='series') and (count(quality(resolution(streams,'1080p'),'Bluray REMUX'))>=1) and count(quality(resolution(streams,'2160p'),'Bluray REMUX')) == 0 and count(quality(resolution(streams,'2160p'),'WEB-DL','WEBRip')) == 0? negate(merge(seadex(streams),library(streams)),resolution(streams,'2160p')):[]
```

ESE Bad 4k Bluray

Used in: All active templates

Excludes 4K BluRay releases below quality thresholds — BR-DISK rips, poorly-encoded remuxes, or flagged problematic groups. Works with ranked regex negative scoring.

```
(queryType=='movie' or queryType=='series') and count(quality(resolution(streams,'2160p'),'Bluray REMUX')) == 0 and count(seadex(resolution(streams,'2160p'))) == 0? negate(merge(seadex(streams),library(streams)),resolution(quality(streams,'Bluray'),'2160p')):[]
```

ESE Bad 1080P Bluray

Used in: All active templates

Mirrors Bad 4k Bluray for 1080p BluRay releases. Catches low-quality BluRay encodes that score negatively in the regex set but weren't caught by quality filtering alone.

```
(queryType=='movie') and count(quality(resolution(streams,'2160p'),'Bluray REMUX'))==0 and count(quality(resolution(streams,'1080p'),'Bluray REMUX'))==0 and count(seadex(resolution(streams,'1080p')))==0? negate(merge(seadex(streams),library(streams)),quality(resolution(streams,'1080p'),'Bluray')):[]
```

ESE Bad NZBs

Used in: All templates with Usenet

Excludes Usenet NZBs flagged as bad — incomplete articles, password-protected archives, or known-broken releases identified by the indexer.

```
merge(releaseGroup(type(streams,'usenet','stremio-usenet'),'sample'),message(type(streams,'usenet','stremio-usenet'),'includes','■'))
```

ESE RD Copyright (per DMM)

Used in: All active templates

Excludes streams flagged for copyright enforcement by RealDebrid via the DMM database. These would fail to resolve on RD even if cached. Kept in all templates for compatibility with hybrid setups.

```
service( keyword(streams, 'all', 'web-dl', 'webrip', 'bdrip', 'hdrip', 'dvdrip', 'bluray.x264', 'hdtv.x264', 'hdtv.xvid', 'web.x264', 'web.h264'), 'realdebrid')
```

ESE Low SEL Score

Used in: Stable (non-Labs) templates

Excludes streams whose stream expression score falls below a minimum threshold after the PSE stack has run. Acts as a final quality floor — catches streams that pass all hard filters but score too poorly to be worth showing. Used in stable templates only; LABS templates replace this with the Score IQR Guard.

```
count(streamExpressionScore(negate(merge(library(streams), seadex(streams)),merge(cached(streams),type(streams), 'p2p', 'http'))), -50))<10?[]:streamExpressionScore(negate(merge(library(streams),seadex(streams)),streams), -1000000, -50)
```

ESE Extra Cached (HQ)

Used in: Stable (non-Labs) templates

Deduplication for high-quality cached streams (BluRay REMUX, BluRay, WEB-DL, WEBRip): at most one result per release group per resolution tier. Uses old 20-clause merge/slice chain. LABS replaces this with perGroup().

```
merge( slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'BluRay REMUX'), '2160p', '1440p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'BluRay REMUX'), '1080p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'BluRay REMUX'), '720p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'BluRay REMUX'), '576p', '480p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'BluRay REMUX'), 'Unknown'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'BluRay'), '2160p', '1440p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'BluRay'), '1080p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'BluRay'), '720p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'BluRay'), '576p', '480p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'BluRay'), 'Unknown'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'WEB-DL'), '2160p', '1440p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'WEB-DL'), '1080p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'WEB-DL'), '720p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'WEB-DL'), '576p', '480p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'WEB-DL'), 'Unknown'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'WEBRip'), '2160p', '1440p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'WEBRip'), '1080p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'WEBRip'), '720p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'WEBRip'), '576p', '480p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams), 'WEBRip'), 'Unknown'), 3))
```

ESE Extra Cached (LQ)

Used in: Stable (non-Labs) templates

Deduplication for lower-quality cached streams (HDTV, HDRip, DVDRip and below): one result per release group.
Old 15-clause chain; LABS replaces with perGroup().

```
merge( slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'HDTV','HDRip'),'2160p','1440p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'DVDRip','HC HD-Rip'),'2160p','1440p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'HDTV','HDRip'),'1080p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'DVDRip','HC HD-Rip'),'1080p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'HDTV','HDRip'),'720p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'DVDRip','HC HD-Rip'),'720p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'HDTV','HDRip','DVDRip','HC HD-Rip'),'576p','480p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'HDTV','HDRip','DVDRip','HC HD-Rip'),'Unknown'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'TC','SCR'),'2160p','1440p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'CAM','TS'),'2160p','1440p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'TC','SCR'),'1080p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'CAM','TS'),'1080p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'CAM','TS','TC','SCR'),'720p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'CAM','TS','TC','SCR'),'576p','480p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'CAM','TS','TC','SCR'),'Unknown'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'Unknown'),'2160p','1440p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'Unknown'),'1080p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'Unknown'),'720p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'Unknown'),'576p','480p'), 3), slice(resolution(quality(negate(merge((library(streams)),uncached(streams)),streams),'Unknown'),'Unknown'), 3))
```

ESE Extra Uncached (All)

Used in: Stable (non-Labs) templates

Deduplication for uncached streams: one result per release group across all tiers. More permissive than cached deduplication. Old 35-clause chain; LABS replaces with perGroup().

```
merge( slice(resolution(quality(uncached(streams),'Bluray REMUX'),'2160p','1440p'), 3),
slice(resolution(quality(uncached(streams),'Bluray REMUX'),'1080p'), 3),
slice(resolution(quality(uncached(streams),'Bluray REMUX'),'720p'), 3),
slice(resolution(quality(uncached(streams),'Bluray REMUX'),'576p','480p'), 3),
slice(resolution(quality(uncached(streams),'Bluray REMUX'),'Unknown'), 3),
slice(resolution(quality(uncached(streams),'Bluray'),'2160p','1440p'), 3),
slice(resolution(quality(uncached(streams),'Bluray'),'1080p'), 3),
slice(resolution(quality(uncached(streams),'Bluray'),'720p'), 3),
slice(resolution(quality(uncached(streams),'Bluray'),'576p','480p'), 3),
slice(resolution(quality(uncached(streams),'Bluray'),'Unknown'), 3),
slice(resolution(quality(uncached(streams),'WEB-DL'),'2160p','1440p'), 3),
slice(resolution(quality(uncached(streams),'WEB-DL'),'1080p'), 3),
slice(resolution(quality(uncached(streams),'WEB-DL'),'720p'), 3),
slice(resolution(quality(uncached(streams),'WEB-DL'),'576p','480p','360p'), 3),
slice(resolution(quality(uncached(streams),'WEB-DL'),'Unknown'), 3),
slice(resolution(quality(uncached(streams),'WEBRip'),'2160p','1440p'), 3),
slice(resolution(quality(uncached(streams),'WEBRip'),'1080p'), 3),
slice(resolution(quality(uncached(streams),'WEBRip'),'720p'), 3),
slice(resolution(quality(uncached(streams),'WEBRip'),'576p','480p','360p'), 3),
slice(resolution(quality(uncached(streams),'WEBRip'),'Unknown'), 3),
slice(resolution(quality(uncached(streams),'HDTV','HDRip','DVDRip','HC HD-Rip'),'2160p','1440p'), 3),
slice(resolution(quality(uncached(streams),'HDTV','HDRip','DVDRip','HC HD-Rip'),'1080p'), 3),
slice(resolution(quality(uncached(streams),'HDTV','HDRip','DVDRip','HC HD-Rip'),'720p'), 3),
slice(resolution(quality(uncached(streams),'HDTV','HDRip','DVDRip','HC HD-Rip'),'576p','480p','360p'), 3),
slice(resolution(quality(uncached(streams),'HDTV','HDRip','DVDRip','HC HD-Rip'),'Unknown'), 3),
slice(resolution(quality(uncached(streams),'CAM','TS','TC','SCR'),'2160p','1440p'), 3),
slice(resolution(quality(uncached(streams),'CAM','TS','TC','SCR'),'1080p'), 3),
slice(resolution(quality(uncached(streams),'CAM','TS','TC','SCR'),'720p'), 3),
slice(resolution(quality(uncached(streams),'CAM','TS','TC','SCR'),'576p','480p','360p'), 3),
slice(resolution(quality(uncached(streams),'CAM','TS','TC','SCR'),'Unknown'), 3),
slice(resolution(quality(uncached(streams),'Unknown'),'2160p','1440p'), 3),
slice(resolution(quality(uncached(streams),'Unknown'),'1080p'), 3),
slice(resolution(quality(uncached(streams),'Unknown'),'720p'), 3),
slice(resolution(quality(uncached(streams),'Unknown'),'576p','480p','360p'), 3),
slice(resolution(quality(uncached(streams),'Unknown'),'Unknown'), 3))
```

ESE Final Limit (All)

Used in: All active templates

Hard cap on total results after all ESEs run. Enforces the template's `maxResults` setting at expression level regardless of how many streams scrapers return.

```
merge(count(quality(negate(merge(library(streams),
seadex(streams)),merge(cached(streams),type(streams,'p2p','http'))),'Bluray REMUX','Bluray','WEB-
DL','WEBRip'))>12? quality(negate(merge(library(streams), seadex(streams)),streams),'HDRip','HC HD-
Rip','DVRip','HDTV','CAM','TS','TC','SCR'):count(quality(negate(merge(library(streams),
seadex(streams)),merge(cached(streams),type(streams,'p2p','http'))),'Bluray REMUX','Bluray','WEB-
DL','WEBRip','HDRip','HC HD-Rip','DVRip','HDTV'))>12? quality(streams,'CAM','TS','TC','SCR'):[]),
count(resolution(negate(merge(library(streams),
seadex(streams)),merge(cached(streams),type(streams,'p2p','http'))),'2160p','1440p','1080p'))>15?
negate(merge(library(streams), seadex(streams)),merge(type(uncached(streams),'debrid'),
resolution(merge(type(streams,'usenet','stremio-
usenet','http','p2p'),type(cached(streams),'debrid')),'720p','576p','480p','360p','240p','144p','Unknown'))):
count(resolution(negate(merge(library(streams),
seadex(streams)),merge(cached(streams),type(streams,'p2p','http'))),'2160p','1440p','1080p'))>12?
negate(merge(library(streams), seadex(streams)),merge(type(uncached(streams),'debrid'),
slice(resolution(merge(type(streams,'usenet','stremio-usenet','http','p2p'),type(cached(streams),'debrid')),'7
20p','576p','480p','360p','240p','144p','Unknown'),3))): count(resolution(negate(merge(library(streams),
seadex(streams)),merge(cached(streams),type(streams,'p2p','http'))),'2160p','1440p','1080p','720p'))>12?
negate(merge(library(streams), seadex(streams)),merge(type(uncached(streams),'debrid'),
resolution(merge(type(streams,'usenet','stremio-
usenet','http','p2p'),type(cached(streams),'debrid')),'576p','480p','360p','240p','144p','Unknown'))):
count(resolution(negate(merge(library(streams),
seadex(streams)),merge(cached(streams),type(streams,'p2p','http'))),'2160p','1440p','1080p','720p'))>9?
negate(merge(library(streams), seadex(streams)),merge(type(uncached(streams),'debrid'),
slice(resolution(merge(type(streams,'usenet','stremio-
usenet','http','p2p'),type(cached(streams),'debrid')),'576p','480p','360p','240p','144p','Unknown'), 3))):
count(resolution(negate(merge(library(streams),
seadex(streams)),merge(cached(streams),type(streams,'p2p','http'))),'2160p','1440p','1080p','720p'))>6?
negate(merge(library(streams), seadex(streams)),merge(type(uncached(streams),'debrid'),
slice(resolution(merge(type(streams,'usenet','stremio-
usenet','http','p2p'),type(cached(streams),'debrid')),'576p','480p','360p','240p','144p','Unknown'),6))):
count(resolution(negate(merge(library(streams), seadex(streams)),
merge(cached(streams),type(streams,'p2p','http'))),'2160p','1440p','1080p','720p'))>3?
negate(merge(library(streams), seadex(streams)),merge(slice(type(uncached(streams),'debrid'), 6),
slice(resolution(merge(type(streams,'usenet','stremio-
usenet','http','p2p'),type(cached(streams),'debrid')),'576p','480p','360p','240p','144p','Unknown'),-1))):[])
```

ESE CB | Hard CAM Kill

Used in: All active templates

Immediately excludes CAM, SCR, TS, TC, and HC HD-Rip streams. Pre-release or cinema-recorded qualities with no use case in a quality-focused template.

```
quality(streams, 'CAM', 'SCR', 'TS', 'TC', 'HC HD-Rip')
```

ESE CB | Hard External Kill

Used in: All active templates

Excludes streams of type 'external' — HTTP redirects that open a browser instead of playing inline in Stremio. They bypass AIOStreams' formatting and filtering entirely.

```
type(streams, 'external')
```

ESE CB | Kill Multi-Episode When Singles Exist

Used in: All active templates

For series queries: if 3+ single-episode results exist, excludes multi-episode packs that aren't full season packs. A poor match when requesting a specific episode.

```
(queryType == 'series' or queryType == 'anime.series') ? (count(negate(streams, multiEpisode(streams))) >= 3 ?
negate(multiEpisode(streams), seasonPack(streams)) : []) : []
```


ESE CB | Kill Season Packs When Episodes Exist

Used in: All active templates

If 3+ individual episode results exist, excludes full season packs. Season packs require manual file selection when you want a specific episode.

```
count(negate(streams, seasonPack(streams))) >= 3 ? seasonPack(streams) : []
```

ESE CB | Kill Ambiguous Packs When Non-Pack Exist

Used in: All active templates

When any non-pack result exists, excludes ambiguous packs (season-like packs without clear episode labels). Prevents vague multi-file archives from surfacing.

```
count(negate(streams, seasonPack(streams))) > 0 ? negate(seasonPack(streams), episodePack(streams)) : []
```

Priority Stream Expressions (PSE)

PSEs run after ESEs and sort the remaining streams into quality tiers. Each PSE returns a subset — the union of all results in order becomes the ranked list. APEX templates use IQR Tukey fence bitrate windowing; CB templates use simpler resolution×quality filters without statistical windowing.

IQR three-tier adaptive logic: ≥ 4 ranked peers $\rightarrow Q1-1.5 \times IQR$ to $Q3+1.5 \times IQR$. 1–3 peers $\rightarrow \min \times 0.8$ to $\max \times 1.2$. 0 peers $\rightarrow \text{median} \times (1 \pm 0.4 \times \text{pow}(0.95, \text{days}))$. Decay: $\pm 40\%$ day 0 $\rightarrow \pm 9\%$ day 30 $\rightarrow \pm 2\%$ day 60 $\rightarrow \sim 0\%$ day 90.

PSE APEX S-Tier 4K Remux — IQR Tukey fence

Used in: 4K Apex, 4K Apex TorBox, 4K Hybrid, 4K Essential, 4K AllDebrid

Top priority for 4K BluRay REMUX. Three-tier adaptive bitrate window: (1) ≥ 4 ranked peers $\rightarrow$ IQR Tukey fence ($Q1-1.5 \times IQR$ to $Q3+1.5 \times IQR$), 15 GB floor; (2) 1–3 peers $\rightarrow \min \times 0.8 - \max \times 1.2$; (3) 0 peers $\rightarrow$ exponential decay: $\text{median} \times (1 \pm 0.4 \times \text{pow}(0.95, \text{daysSinceRelease}))$. The decay gives $\pm 40\%$ on day 0, $\pm 9\%$ at day 30, $\pm 2\%$ at day 60, $\sim 0\%$ at day 90. New releases get a wide window that tightens as the 'correct' bitrate establishes.

```
count(resolution(quality(streams, 'BluRay REMUX'), '2160p'))>=4?size(bitrate(resolution(quality(streams, 'BluRay REMUX'), '2160p'), q1(values(resolution(quality(streams, 'BluRay REMUX'), '2160p'), 'bitrate'))-1.5*iqr(values(resolution(quality(streams, 'BluRay REMUX'), '2160p'), 'bitrate')), q3(values(resolution(quality(streams, 'BluRay REMUX'), '2160p'), 'bitrate'))+1.5*iqr(values(resolution(quality(streams, 'BluRay REMUX'), '2160p'), 'bitrate'))), '15GB'):count(resolution(quality(streams, 'BluRay REMUX'), '2160p'))>0?size(bitrate(resolution(quality(streams, 'BluRay REMUX'), '2160p'), min(values(resolution(quality(streams, 'BluRay REMUX'), '2160p'), 'bitrate'))*0.80, max(values(resolution(quality(streams, 'BluRay REMUX'), '2160p'), 'bitrate'))*1.20), '15GB'):[]
```

PSE APEX A-Tier 4K WEB-DL HDR

Used in: 4K Apex, 4K Hybrid, 4K Essential, 4K AllDebrid

4K WEB-DL with HDR (DV, HDR10+, HDR10, HDR, HDR+DV). Same three-tier adaptive window. HDR required — SDR 4K WEB-DL falls to B-Tier.

```
count(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'))>=4?bitrate(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), q1(values(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate'))-1.5*iqr(values(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate')), q3(values(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate'))+1.5*iqr(values(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate'))):count(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'))>0?bitrate(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), min(values(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate'))*0.80, max(values(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate'))*1.20):((count(bitrate(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), median(values(bitrate(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), '5Mbps'), 'bitrate'))*(1-0.4*pow(0.95, daysSinceRelease)), median(values(bitrate(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), '5Mbps'), 'bitrate'))*(1+0.4*pow(0.95, daysSinceRelease))))>=1?bitrate(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), median(values(bitrate(resolution(visualTag(quality(streams, 'WEB-DL'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), '5Mbps'), 'bitrate'))*(1-0.4*pow(0.95, daysSinceRelease))):[]
```

PSE APEX B-Tier 4K WEB-DL SDR

Used in: 4K Apex, 4K Hybrid, 4K Essential, 4K AllDebrid

4K WEB-DL without HDR. Rare — most 4K WEB-DL is HDR-mastered. Still uses adaptive window.

```
count(resolution(quality(streams, 'WEB-DL'), '2160p'))>=4?bitrate(resolution(quality(streams, 'WEB-DL'), '2160p'), q1(values(resolution(quality(streams, 'WEB-DL'), '2160p'), 'bitrate'))-1.5*iqr(values(resolution(quality(streams, 'WEB-DL'), '2160p'), 'bitrate')), q3(values(resolution(quality(streams, 'WEB-DL'), '2160p'), 'bitrate'))+1.5*iqr(values(resolution(quality(streams, 'WEB-DL'), '2160p'), 'bitrate'))):count(resolution(quality(streams, 'WEB-DL'), '2160p'))>0?bitrate(resolution(quality(streams, 'WEB-DL'), '2160p'), min(values(resolution(quality(streams, 'WEB-DL'), '2160p'), 'bitrate'))*0.80, max(values(resolution(quality(streams, 'WEB-DL'), '2160p'), 'bitrate'))*1.20):((count(bitrate(resolution(quality(streams, 'WEB-DL'), '2160p'), median(values(bitrate(resolution(quality(streams, 'WEB-DL'), '2160p'), '5Mbps'), 'bitrate'))*(1-0.4*pow(0.95, daysSinceRelease)), median(values(bitrate(resolution(quality(streams, 'WEB-DL'), '2160p'), '5Mbps'), 'bitrate'))*(1+0.4*pow(0.95, daysSinceRelease))))>=1?bitrate(resolution(quality(streams, 'WEB-DL'), '2160p'), median(values(bitrate(resolution(quality(streams, 'WEB-DL'), '2160p'), '5Mbps'), 'bitrate'))*(1-0.4*pow(0.95, daysSinceRelease))):[]))
```

PSE APEX C-Tier 4K WEBRip HDR

Used in: 4K Apex, 4K Hybrid, 4K Essential, 4K AllDebrid

4K WEBRip with HDR. WEBRip is a re-encode of WEB-DL rather than a direct stream extraction — slightly lower quality. Adaptive window applies.

```
count(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'))>=4?bitrate(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), q1(values(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate'))-1.5*iqr(values(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate')), q3(values(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate'))+1.5*iqr(values(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate'))):count(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'))>0?bitrate(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), min(values(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate'))*0.80, max(values(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), 'bitrate'))*1.20):((count(bitrate(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), median(values(bitrate(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), '5Mbps'), 'bitrate'))*(1-0.4*pow(0.95, daysSinceRelease)), median(values(bitrate(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), '5Mbps'), 'bitrate'))*(1+0.4*pow(0.95, daysSinceRelease))))>=1?bitrate(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), median(values(bitrate(resolution(visualTag(quality(streams, 'WEBRip'), 'HDR+DV', 'DV', 'HDR10+', 'HDR10', 'HDR'), '2160p'), '5Mbps'), 'bitrate'))*(1-0.4*pow(0.95, daysSinceRelease))):[]))
```

PSE APEX D-Tier 4K WEBRip SDR

Used in: 4K Apex, 4K Hybrid, 4K Essential, 4K AllDebrid

4K WEBRip without HDR. Lowest dedicated 4K tier before the catch-all fallback.

```
count(resolution(quality(streams, 'WEBRip'), '2160p'))>=4?bitrate(resolution(quality(streams, 'WEBRip'), '2160p'), q1(values(resolution(quality(streams, 'WEBRip'), '2160p'), 'bitrate'))-1.5*iqr(values(resolution(quality(streams, 'WEBRip'), '2160p'), 'bitrate')), q3(values(resolution(quality(streams, 'WEBRip'), '2160p'), 'bitrate'))+1.5*iqr(values(resolution(quality(streams, 'WEBRip'), '2160p'), 'bitrate'))):count(resolution(quality(streams, 'WEBRip'), '2160p'))>0?bitrate(resolution(quality(streams, 'WEBRip'), '2160p'), min(values(resolution(quality(streams, 'WEBRip'), '2160p'), 'bitrate'))*0.80, max(values(resolution(quality(streams, 'WEBRip'), '2160p'), 'bitrate'))*1.20):[]
```

PSE APEX E-Tier 4K | Any 2160p fallback

Used in: 4K Apex, 4K Hybrid, 4K Essential, 4K AllDebrid

Catch-all: any 2160p stream that missed all upper tiers. No bitrate windowing. Common for niche or obscure content with non-standard release types.

```
resolution(streams, '2160p')
```

PSE APEX S-Tier 1080p Remux

Used in: 4K Apex, 4K Hybrid, 4K Essential, 4K AllDebrid

1080p BluRay REMUX fallback when no 4K passes upper tiers. Three-tier adaptive window, 8 GB size floor (vs 15 GB for 4K REMUX).

```
count(resolution(quality(streams, 'BluRay REMUX'), '1080p'))>=4?size(bitrate(resolution(quality(streams, 'BluRay REMUX'), '1080p'), q1(values(resolution(quality(streams, 'BluRay REMUX'), '1080p'), 'bitrate'))-1.5*iqr(values(resolution(quality(streams, 'BluRay REMUX'), '1080p'), 'bitrate'))), q3(values(resolution(quality(streams, 'BluRay REMUX'), '1080p'), 'bitrate'))+1.5*iqr(values(resolution(quality(streams, 'BluRay REMUX'), '1080p'), 'bitrate'))), '8GB'):count(resolution(quality(streams, 'BluRay REMUX'), '1080p'))>0?size(bitrate(resolution(quality(streams, 'BluRay REMUX'), '1080p'), min(values(resolution(quality(streams, 'BluRay REMUX'), '1080p'), 'bitrate'))*0.80, max(values(resolution(quality(streams, 'BluRay REMUX'), '1080p'), 'bitrate'))*1.20), '8GB'):[]
```

PSE APEX A-Tier 1080p WEB-DL

Used in: 4K Apex, 4K Hybrid, 4K Essential, 4K AllDebrid

1080p WEB-DL fallback. Adaptive window applied.

```
count(resolution(quality(streams, 'WEB-DL'), '1080p'))>=4?bitrate(resolution(quality(streams, 'WEB-DL'), '1080p'), q1(values(resolution(quality(streams, 'WEB-DL'), '1080p'), 'bitrate'))-1.5*iqr(values(resolution(quality(streams, 'WEB-DL'), '1080p'), 'bitrate'))), q3(values(resolution(quality(streams, 'WEB-DL'), '1080p'), 'bitrate'))+1.5*iqr(values(resolution(quality(streams, 'WEB-DL'), '1080p'), 'bitrate'))):count(resolution(quality(streams, 'WEB-DL'), '1080p'))>0?bitrate(resolution(quality(streams, 'WEB-DL'), '1080p'), min(values(resolution(quality(streams, 'WEB-DL'), '1080p'), 'bitrate'))*0.80, max(values(resolution(quality(streams, 'WEB-DL'), '1080p'), 'bitrate'))*1.20):(((count(bitrate(resolution(quality(streams, 'WEB-DL'), '1080p'), median(values(bitrate(resolution(quality(streams, 'WEB-DL'), '1080p'), '1Mbps'), 'bitrate'))*(1-0.4*pow(0.95, daysSinceRelease)), median(values(bitrate(resolution(quality(streams, 'WEB-DL'), '1080p'), '1Mbps'), 'bitrate'))*(1+0.4*pow(0.95, daysSinceRelease))))>=1?bitrate(resolution(quality(streams, 'WEB-DL'), '1080p'), median(values(bitrate(resolution(quality(streams, 'WEB-DL'), '1080p'), '1Mbps'), 'bitrate'))*(1-0.4*pow(0.95, daysSinceRelease)))):[]))
```

PSE APEX B-Tier 1080p | WEBRip or BluRay

Used in: All templates with 1080p tier

1080p WEBRip or BluRay encode. Simple quality×resolution filter; no bitrate window.

```
resolution(quality(streams, 'WEBRip', 'BluRay'), '1080p')
```

PSE APEX C-Tier 1080p | Any 1080p

Used in: All templates with 1080p tier

Catch-all 1080p — any stream at 1080p that missed higher tiers.

```
resolution(streams, '1080p')
```

PSE APEX 720p | WEB-DL or WEBRip

Used in: All templates with 720p fallback

720p fallback for WEB-DL or WEBRip. Fires only when no 1080p results pass.

```
resolution(quality(streams, 'WEB-DL', 'WEBRip'), '720p')
```

PSE APEX 720p | Any

Used in: All templates with 720p fallback

Final catch-all: any 720p stream. Last resort before showing zero results.

```
resolution(streams, '720p')
```

PSE Codec Efficiency Booster

Used in: All active templates

Within the same quality+resolution tier, boosts HEVC and AV1 encodes over AVC. Modern codecs deliver equivalent quality at 30–50% lower bitrate — faster start and lower data use. Applied to cached non-library non-SeaDex streams at 1080p and 720p.

```
merge(encode(resolution(quality(negate(merge(library(streams),seadex(streams)),cached(streams)), 'Bluray REMUX'), '1080p'), 'HEVC', 'AV1'), encode(resolution(quality(negate(merge(library(streams),seadex(streams)),cached(streams)), 'WEB-DL', 'WEBRip'), '1080p'), 'HEVC', 'AV1'), encode(resolution(quality(negate(merge(library(streams),seadex(streams)),cached(streams)), 'Bluray REMUX'), '720p'), 'HEVC', 'AV1'), encode(resolution(quality(negate(merge(library(streams),seadex(streams)),cached(streams)), 'WEB-DL', 'WEBRip'), '720p'), 'HEVC', 'AV1'))
```

PSE Audio Pinnacle

Used in: All active templates

Boosts premium audio: first Atmos/TrueHD, then DTS-HD MA/DTS-X, then EAC3/DD+. Applied to cached non-library non-SeaDex streams. Ensures best audio mix surfaces rather than relying on scraper order.

```
merge(audioTag(negate(merge(library(streams),seadex(streams)),cached(streams)), 'Atmos', 'TrueHD'), audioTag(negate(merge(library(streams),seadex(streams)),cached(streams)), 'DTS-HD MA', 'DTS-X'), audioTag(negate(merge(library(streams),seadex(streams)),cached(streams)), 'EAC3', 'DD+'))
```

PSE HDR/DV Priority

Used in: All templates with 4K tier

Within 4K results, boosts DV, HDR10+, and HDR10+/DV above HDR10 and HDR. Applied to cached 4K non-library non-SeaDex streams. DV profile 8 and HDR10+ are dynamically tone-mapped on supported displays.

```
merge(visualTag(resolution(cached(negate(merge(library(streams),seadex(streams)),streams)), '2160p'), 'DV', 'HDR10+', 'HDR10+/DV'), visualTag(resolution(cached(negate(merge(library(streams),seadex(streams)),streams)), '2160p'), 'HDR10', 'HDR'))
```

PSE Boost Cached Usenet

Used in: All active templates

Boosts cached Usenet streams, library results, and SeaDex results to the top. Usenet delivers near-instant playback (no seeder dependency) and library/SeaDex results are pre-validated. Ensures these source types surface above cached torrents.

```
merge(cached(merge(library(streams),seadex(streams),type(streams, 'debrid', 'usenet', 'stremio-usenet'))))
```

Inclusion Stream Expressions (ISE)

ISEs run last and re-insert streams that ESEs may have removed. They act as safety nets: ensuring library content, fallback streams, and REPACK/PROPER releases are never silently dropped by aggressive exclusion logic.

ISE Part of Tamtaro SEL Setup

Used in: All active templates

Placeholder ISE grouping marker. Returns `[]` always. Real ISE logic loads from `syncdIncludedStreamExpressionUrls`.

```
[]
```

ISE ISE v1.2.3* [git.tamtaro.de]

Used in: All active templates

Version label for Tamtaro's inclusion set. Returns `[]` inline; syncd URL does the work.

```
[]
```

ISE Library

Used in: All active templates

Passes library (continue-watching) streams through when all result streams are library streams. Ensures in-progress episodes are never dropped by ESEs.

```
count(streams)==count(library(streams))?library(streams):[]
```

ISE digitalRelease Bypass

Used in: All active templates

For movies/anime movies: allows CAM-quality streams through during the blackout window before the official digital release. Prevents zero results on brand-new theatrical releases that haven't had a digital rip yet.

```
queryType=='movie' or queryType=='anime.movie'?  
(count(passthrough(quality(streams,'CAM','TS','TC','SCR','WEBRip'),'digitalRelease')) > 15 ?  
passthrough(quality(streams,'CAM','TS','TC','SCR','WEBRip'),'digitalRelease') :  
passthrough(streams,'digitalRelease')) : []
```

ISE 0Cached

Used in: All active templates

When zero cached results exist across all source types (debrid, p2p, HTTP, Usenet, Stremio-Usenet), passes all streams through as a last resort. Showing uncached results is always better than showing nothing.

```
count(merge(cached(streams), type(streams, 'p2p','http','usenet','stremio-usenet')))==0?passthrough(streams,'title'):[]
```

ISE REPACK/PROPER Passthrough

Used in: All active templates

Forces REPACK and PROPER releases through regardless of other filtering. If a REPACK has at least one cached result, it surfaces unconditionally — REPACKs fix defects in the original release and should always take priority over the defective source.

```
count(keyword(negate(merge(library(streams),seadex(streams)),streams),'all','repack','proper'))>0?passthrough(  
keyword(negate(merge(library(streams),seadex(streams)),streams),'all','repack','proper'),'excluded','limit'):[]
```

LABS Expressions (v0.9.0 — Nightly)

LABS expressions are in testing in the Nightly template tier (4K Apex Labs, Anime 4K Labs, All-Rounder Labs). They graduate to the full stable suite after a 2–3 week testing window if no regressions are reported.

LABS ESEs

LABS LABS v0.9.0 | Score IQR Guard

Used in: LABS templates (4K Apex Labs, Anime 4K Labs, All-Rounder Labs)

Adaptive low-score filter using the IQR Tukey fence on stream expression scores. When 8+ non-library, non-SeaDex cached/p2p/HTTP streams exist, excludes any stream whose seScore falls below $Q1 - 1.5 \times IQR$ of the field. The 8-stream minimum prevents the guard from firing on thin result sets where the IQR is meaningless. Replaces the flat 'Low SEL Score' ESE with a statistically adaptive threshold that scales with the actual result quality distribution.

```
count(negate(merge(library(streams), seadex(streams)), merge(cached(streams), type(streams, 'p2p', 'http')))) >=
8 ? streamExpressionScore(negate(merge(library(streams), seadex(streams)), streams), -1000000,
q1(values(negate(merge(library(streams), seadex(streams)), streams), 'seScore')) - 1.5 *
iqr(values(negate(merge(library(streams), seadex(streams)), streams), 'seScore')) : []
```

LABS LABS v0.9.0 | Bad Dual Audio Groups

Used in: LABS templates

Hard-excludes known dual-audio release groups via releaseGroup(): alfaHD, BAT, BiOMA, BlackBit, BNd, Cory, EXTREME, FF, FOXX, G4RiS, GUEIRA, LCD, N3G4N, PD, PTHome, RiPER, RK, SiGLA, Tars, TM, tokar86a, TURG, vnlls, WTV, Yatogam1, YusukeFLA, ZigZag, ZNM. Replaces the old -50 ranked regex score penalty for these groups. A -50 ranked entry can still surface at position 2 or 3 on thin results; an ESE exclusion removes them entirely regardless of result count.

```
releaseGroup(streams, 'alfaHD', 'BAT', 'BiOMA', 'BlackBit', 'BNd', 'Cory', 'EXTREME', 'FF', 'FOXX', 'G4RiS',
'GUEIRA', 'LCD', 'N3G4N', 'PD', 'PTHome', 'RiPER', 'RK', 'SiGLA', 'Tars', 'TM', 'tokar86a', 'TURG', 'vnlls',
'WTV', 'Yatogam1', 'YusukeFLA', 'ZigZag', 'ZNM')
```

LABS LABS v0.9.0 | Extra Cached HQ — perGroup()

Used in: LABS templates

perGroup() deduplication for high-quality cached streams (BluRay REMUX, BluRay, WEB-DL, WEBRip): one result per release group per resolution, capped at 3. Replaces the old 20-clause merge/slice chain with a single readable expression. Same deduplication behaviour, ~90% less expression code.

```
count(negate(merge(library(streams), seadex(streams)), merge(cached(streams), type(streams, 'p2p', 'http')))) >=
8 ? streamExpressionScore(negate(merge(library(streams), seadex(streams)), streams), -1000000,
q1(values(negate(merge(library(streams), seadex(streams)), streams), 'seScore')) - 1.5 *
iqr(values(negate(merge(library(streams), seadex(streams)), streams), 'seScore')) : []
```

LABS LABS v0.9.0 | Extra Cached LQ — perGroup()

Used in: LABS templates

perGroup() deduplication for low-quality cached streams (HDTV, HDRip, DVDRip, HC HD-Rip, CAM, TS, TC, SCR, Unknown): one per group per resolution, cap 3. Replaces the 15-clause chain.

```
count(negate(merge(library(streams), seadex(streams)), merge(cached(streams), type(streams, 'p2p', 'http')))) >=
8 ? streamExpressionScore(negate(merge(library(streams), seadex(streams)), streams), -1000000,
q1(values(negate(merge(library(streams), seadex(streams)), streams), 'seScore')) - 1.5 *
iqr(values(negate(merge(library(streams), seadex(streams)), streams), 'seScore')) : []
```

LABS LABS v0.9.0 | Extra Uncached — perGroup()

Used in: LABS templates

perGroup() deduplication for all uncached streams: one per release group per resolution, cap 3. Replaces the 35-clause chain. Simplest and most impactful of the three perGroup() replacements.

```
count(negate(merge(library(streams), seadex(streams)), merge(cached(streams), type(streams, 'p2p', 'http')))) >= 8 ? streamExpressionScore(negate(merge(library(streams), seadex(streams)), streams), -1000000, q1(values(negate(merge(library(streams), seadex(streams)), streams), 'seScore')) - 1.5 * iqr(values(negate(merge(library(streams), seadex(streams)), streams), 'seScore')) : []
```

LABS LABS v0.9.0 | Indexer Diversity

Used in: LABS templates

When 20+ non-library, non-SeaDex cached/p2p/HTTP streams exist, caps results at 2 per scraper source using perGroup('indexer', 2). Prevents any single indexer from filling 8 of your 10 results on popular titles where Comet and MediaFusion both return the same 5 groups. No visible effect on niche titles with thin results.

```
count(negate(merge(library(streams), seadex(streams)), merge(cached(streams), type(streams, 'p2p', 'http')))) > 20 ? negate(perGroup(negate(merge(library(streams), seadex(streams)), merge(cached(streams), type(streams, 'p2p', 'http'))), 'indexer', 2), negate(merge(library(streams), seadex(streams)), merge(cached(streams), type(streams, 'p2p', 'http')))) : []
```

LABS PSEs

LABS LABS v0.9.0 | 4K Remux elite pin

Used in: LABS templates with 4K tier

Pins known-excellent 4K REMUX release groups to the top of results using pin(). Groups: FraMeSToR, DON, FLUX, HIFI, playBD, BMF, QxR, EPSiLON, BLURAniUM, PmP. Pins are applied before IQR PSEs — the bitrate window still runs, but pinned results are guaranteed top placement within their bitrate window tier.

```
pin(releaseGroup(quality(resolution(streams, 'BluRay REMUX'), '2160p'), 'FraMeSToR', 'DON', 'FLUX', 'HIFI', 'playBD', 'BMF', 'QxR', 'EPSiLON', 'BLURAniUM', 'PmP'), 'top')
```

LABS LABS v0.9.0 | 1080p Remux elite pin

Used in: LABS templates with 1080p tier

Pins known-excellent 1080p REMUX groups to the top: NTb, FLUX, KiNGS, NTG, BHDStudio, FraMeSToR, SiC, 126811.

```
pin(releaseGroup(quality(resolution(streams, 'BluRay REMUX'), '2160p'), 'FraMeSToR', 'DON', 'FLUX', 'HIFI', 'playBD', 'BMF', 'QxR', 'EPSiLON', 'BLURAniUM', 'PmP'), 'top')
```

LABS LABS v0.9.0 | LQ pin bottom

Used in: LABS templates

Pins known low-quality groups to the bottom of results: YIFY, RARBG, EVO, YTS, PSA, McGusta, Tigole. These groups consistently produce small, over-compressed files with poor encode quality. Pinning to bottom is more reliable than relying on negative regex scores, which can still surface them at position 2-3 on thin results.

```
pin(releaseGroup(quality(resolution(streams, 'BluRay REMUX'), '2160p'), 'FraMeSToR', 'DON', 'FLUX', 'HIFI', 'playBD', 'BMF', 'QxR', 'EPSiLON', 'BLURAniUM', 'PmP'), 'top')
```

LABS LABS v0.9.0 | IMAX pin

Used in: LABS templates

When any IMAX-tagged stream exists, pins it to the top. IMAX releases use the expanded 1.43:1 or 1.90:1 aspect ratio for IMAX-certified content. Only fires when at least one IMAX stream is present in the result set.

```
count(visualTag(streams, 'IMAX')) > 0 ? pin(visualTag(streams, 'IMAX'), 'top') : []
```


Architecture Notes

Synced vs Inline

Core Builds uses inline expressions (in the template JSON) and synced URL expressions (loaded at runtime from external JSON files). Tamtaro's ESE/ISE sets are synced — they auto-update without a template release. APEX PSEs and CB ESEs are inline, versioned with the template.

Evaluation Order

ESEs → PSEs → ISEs. Within each type, expressions run in list order. Deduplication ESEs run near the end of the ESE list (after quality gates). Final Limit runs last.

APEX vs CB PSE Style

APEX templates use IQR Tukey fence PSEs with statistical bitrate windowing. CB-style templates (Essential Lite, Speed, EasyNews, AllDebrid Lite) use simpler resolution×quality filters — faster and more predictable on thin result sets. IQR requires ≥ 4 ranked streams; CB always fires.

perGroup() Deduplication (LABS)

LABS replaces 20/15/35-clause merge/slice deduplication chains with single perGroup() calls. Identical behaviour: one result per release group per resolution tier. Template size reduction ~4 KB per template.

Flood Guard Caps

Caps apply to non-library non-SeaDex cached streams only. A scraper contributing 20 results gives at most its cap to the final pool. Library and SeaDex bypass all caps.

Score IQR Guard vs Low SEL Score (LABS)

Low SEL Score uses a flat minimum score threshold. Score IQR Guard computes $Q1 - 1.5 \times IQR$ from the actual score distribution — the threshold adapts to whatever the current result pool looks like. Requires ≥ 8 streams to fire; protects thin result sets.

Pin() in LABS

pin(streams, 'top'/'bottom') registers streams in a side-effect map; reordering happens post-evaluation. Used in PSE context — returns [] so streams are not excluded. Pins are applied before IQR PSEs but within the bitrate window tier.